



# 4Arms – Colles Distal Forearm Fracture Reduction Model

C. Gillard, C. Hanby, A. He, R. Rohr

Pratt School of Engineering, Duke University, Duke School of Medicine



## Problem Statement

Hospitals lack effective tools to train residents on treating one of the most common wrist fractures, a distal radius fracture (Figures 1 & 2). Problems with existing models are that they are expensive, fragile, and inaccurate, discouraging hospital investment.

The team aims to develop a realistic, durable, and safe model for training closed reductions at the Duke Hospital.



Figure 1: X-Ray of a healthy wrist.



Figure 2: X-Ray of a fractured distal radius in the wrist.

## Existing Prototype

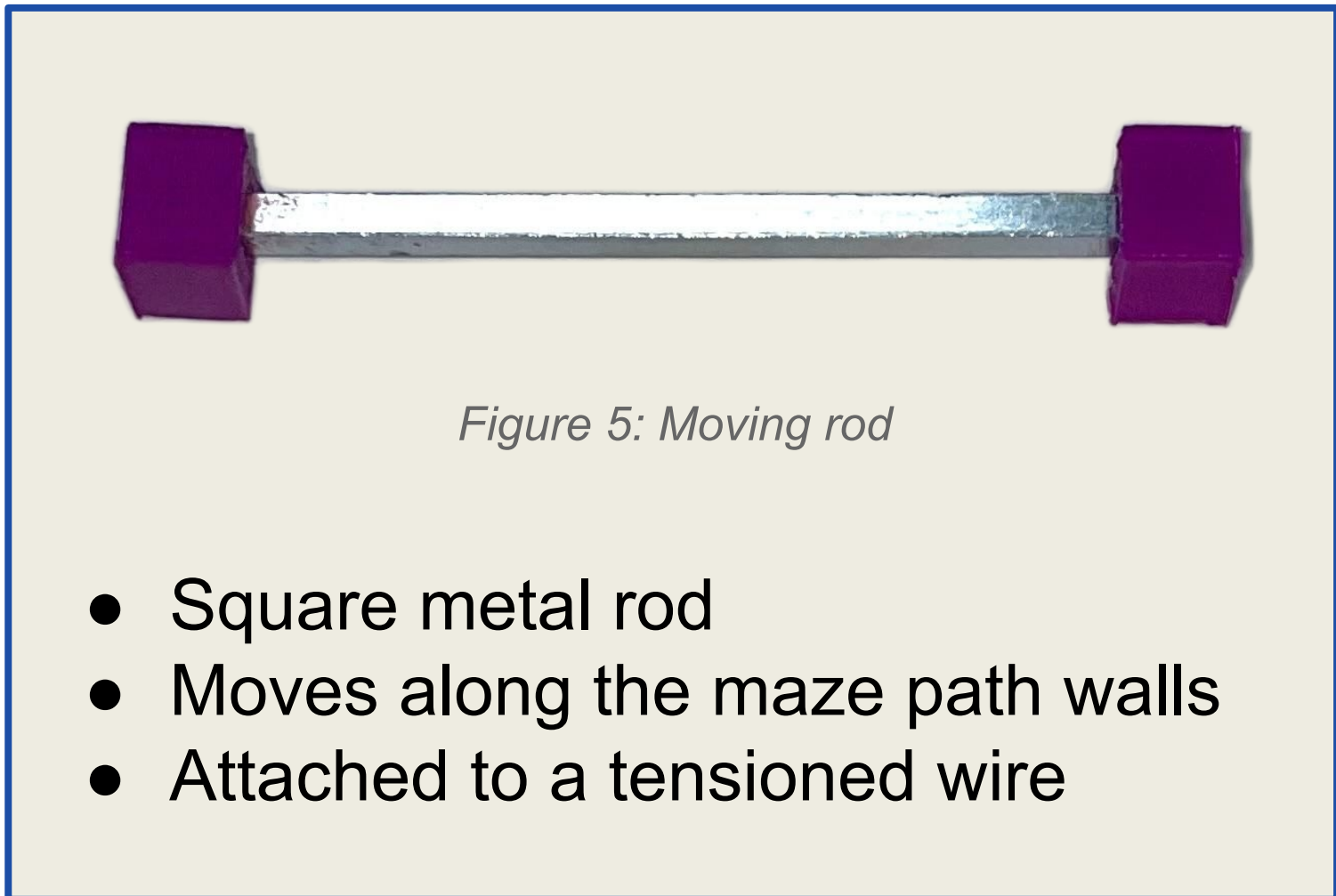


Figure 3: Existing prototype created by engineer David Mercer

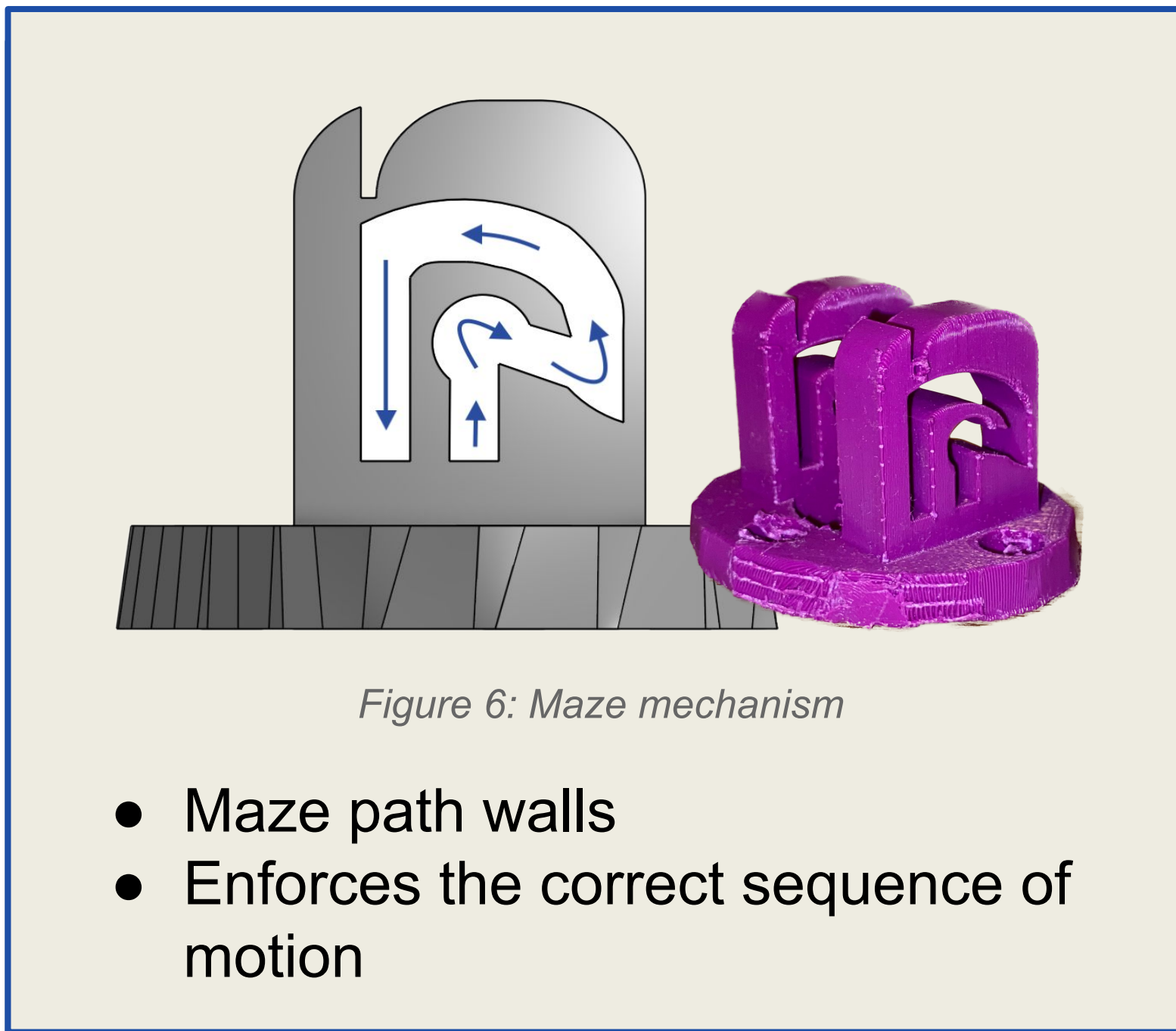
An existing prototype features a 3D-printed hand and forearm, a spring to replicate forces, and a "step" mechanism to mimic the fracture. Issues to address include:

- Unrealistic horizontal movement
- Does not enforce correct sequence of motion
- Overall lack of realism

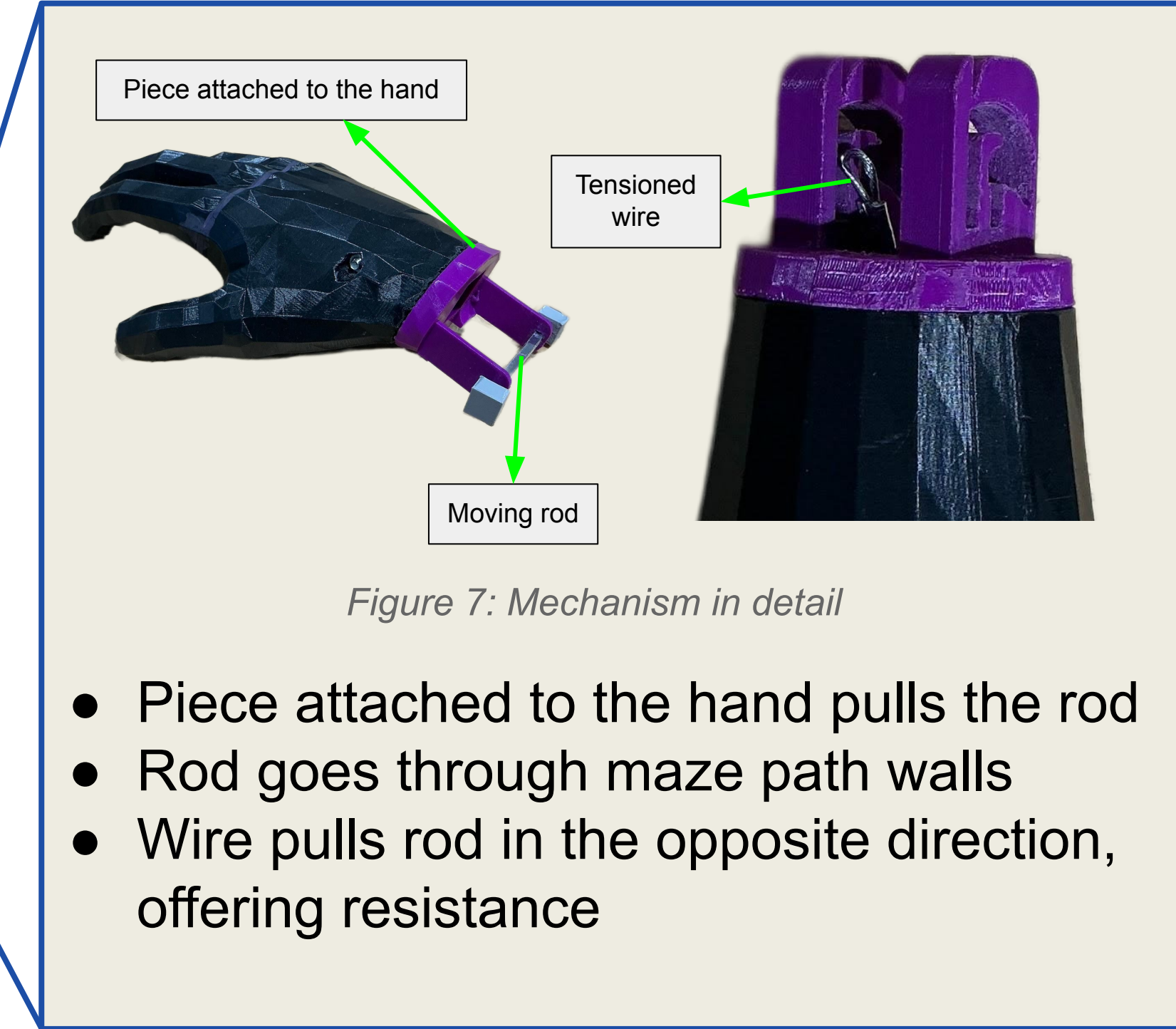
## Final Prototype Model



- Square metal rod
- Moves along the maze path walls
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- Maze path walls
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- Piece attached to the hand pulls the rod
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- Wire pulls rod in the opposite direction, offering resistance



- Removable neoprene sleeve
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Figure 4: Final Prototype with visible attached mechanism

## Design Criteria and Testing

| Design Criteria             | Procedure                                                                                                                                                           | Target Value                                                | Results                                 |
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| Accurate Mechanism & Forces | 8 physicians performed 10 reductions each using the arm and were asked to rate the accuracy of the force and of the reduction movements on a 10-point Likert Scale. | $\geq 8.0/10$ Average Motion<br>$\geq 8.0/10$ Average Force | Motion: Pass (8.3)<br>Force: Pass (8.1) |
| Safety                      | 20 students used the device and reported on the safety perceptions on a 5-point Likert scale.                                                                       | $\geq 4.5/5$ Average                                        | Pass (4.9)                              |
| Durability                  | The full process of reduction and reset were repeated by team members, students, and physicians to assess the frequency or ease of damage on a 3-point scale.       | $\geq 2.8/3$ Average                                        | Pass (2.8)                              |
| Easy Reset                  | 20 undergraduate students were asked to reset the reduced arm while being timed, in addition to answering a user-defined scale.                                     | $< 5$ seconds<br>$\geq 2.5/3$ Average                       | Fail (10.3 seconds)<br>Pass (2.5)       |
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## Conclusion & Future Work

The team's design was successful, achieving its main objective of accurately replicating a distal forearm reduction maneuver, in addition to the approval of the mechanism and forces by physicians. However, there remains room for improvement. Moving forward the group aims to:

- Design an automated mechanism for force calibration and adjustment.
- Make the model more realistic with materials like silicone.
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Figure 9: Final prototype being used by resident at the Duke Hospital.

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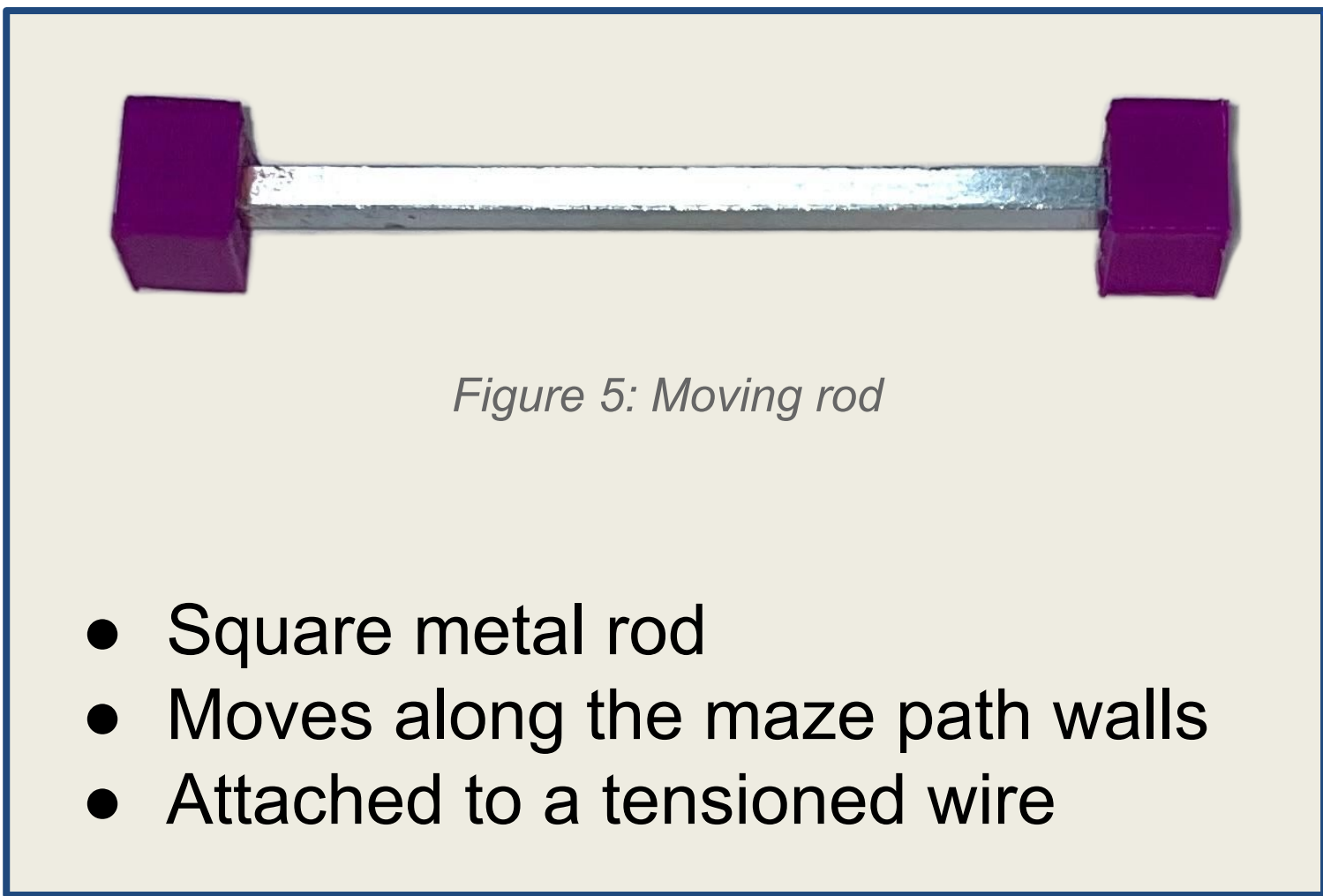


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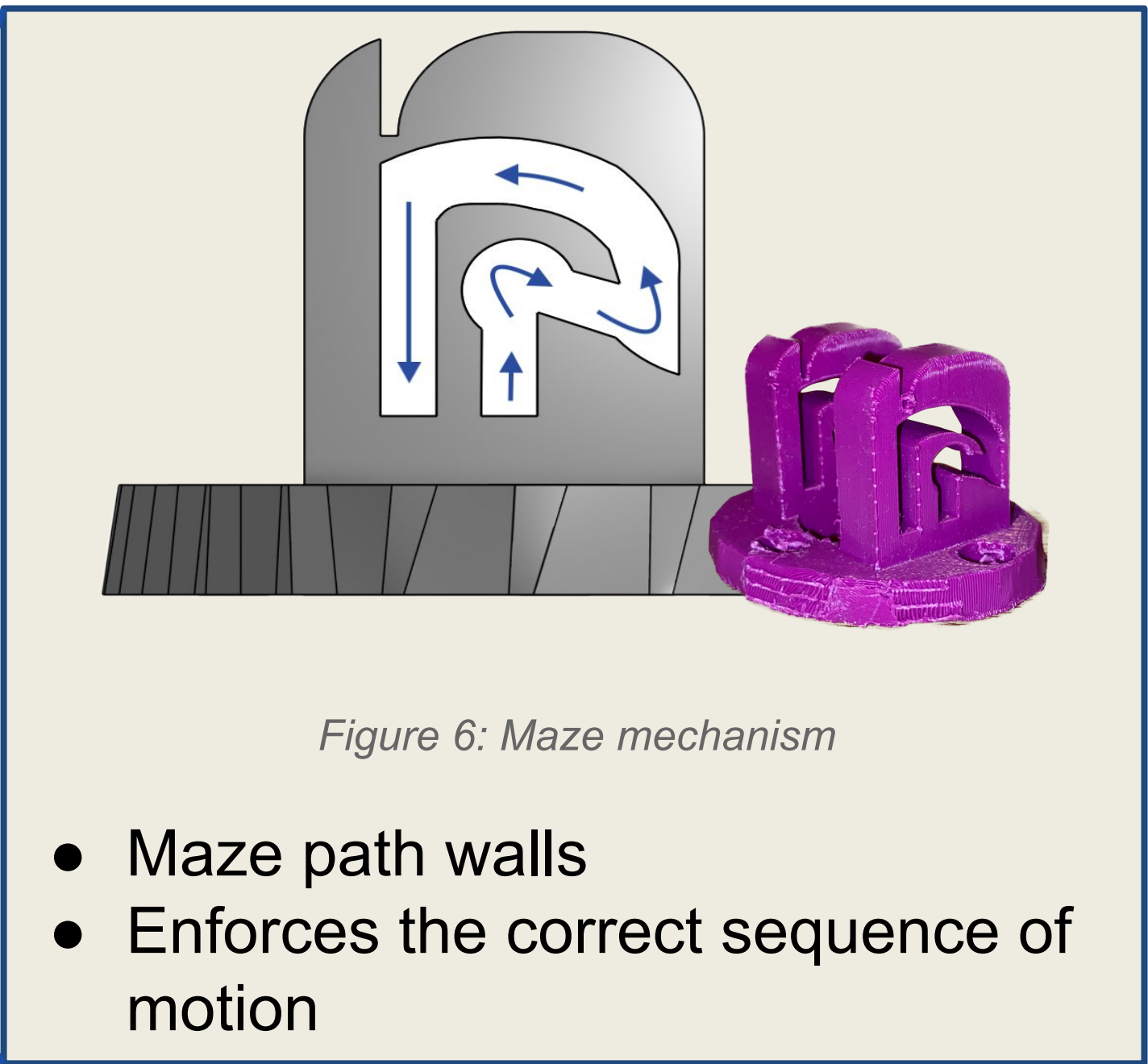


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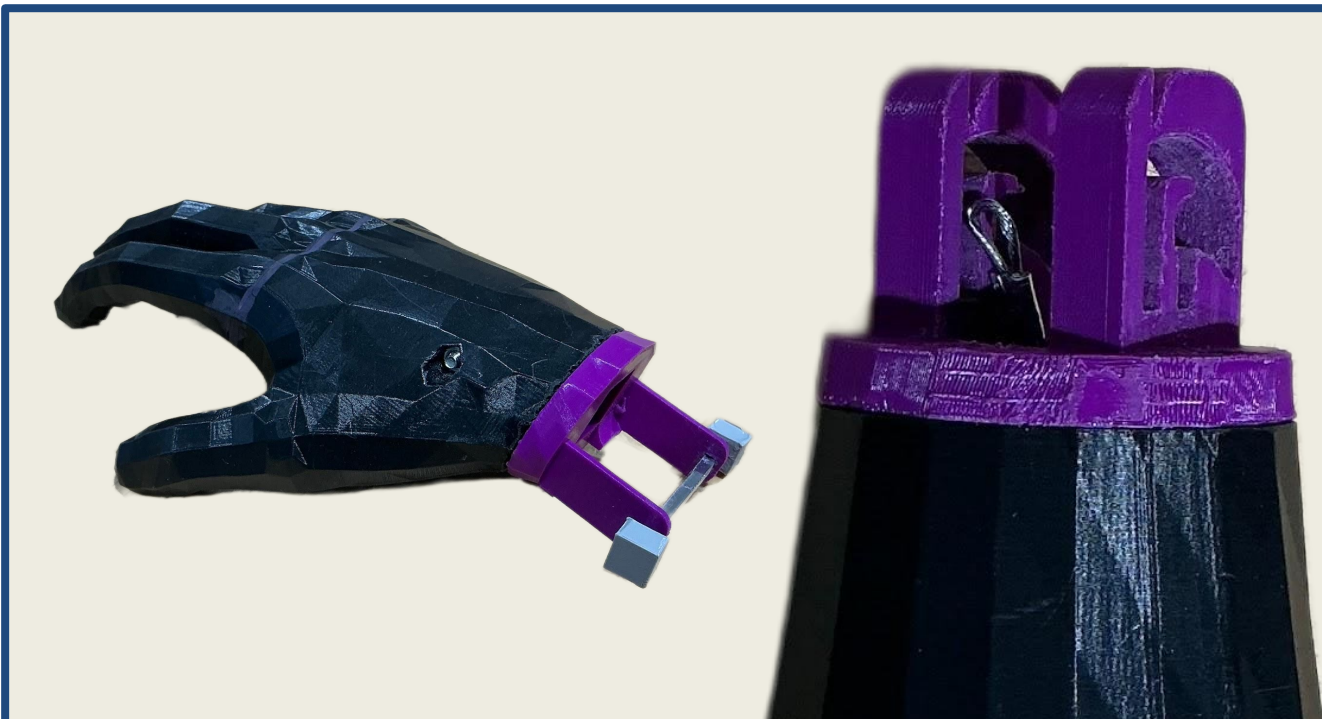


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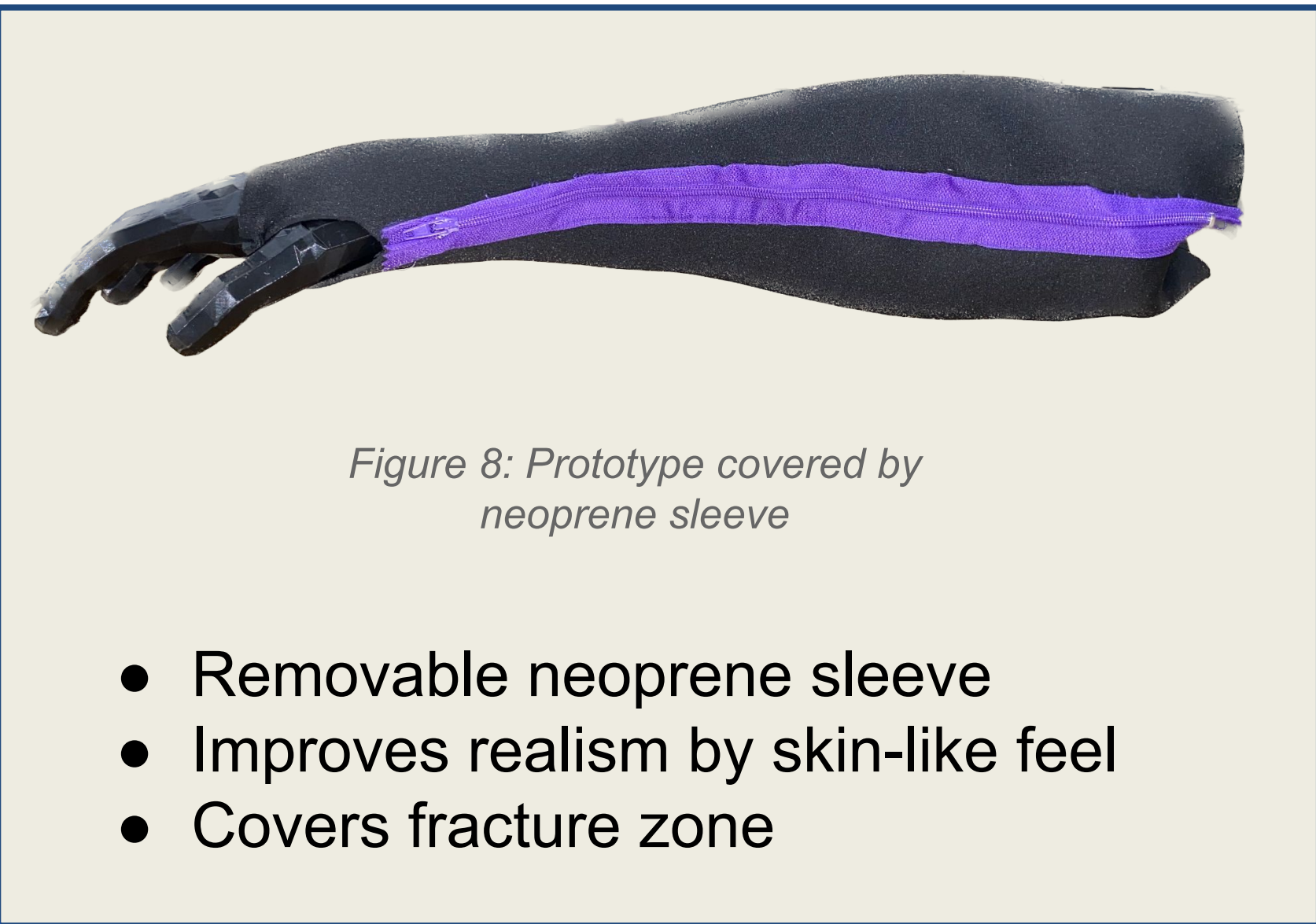


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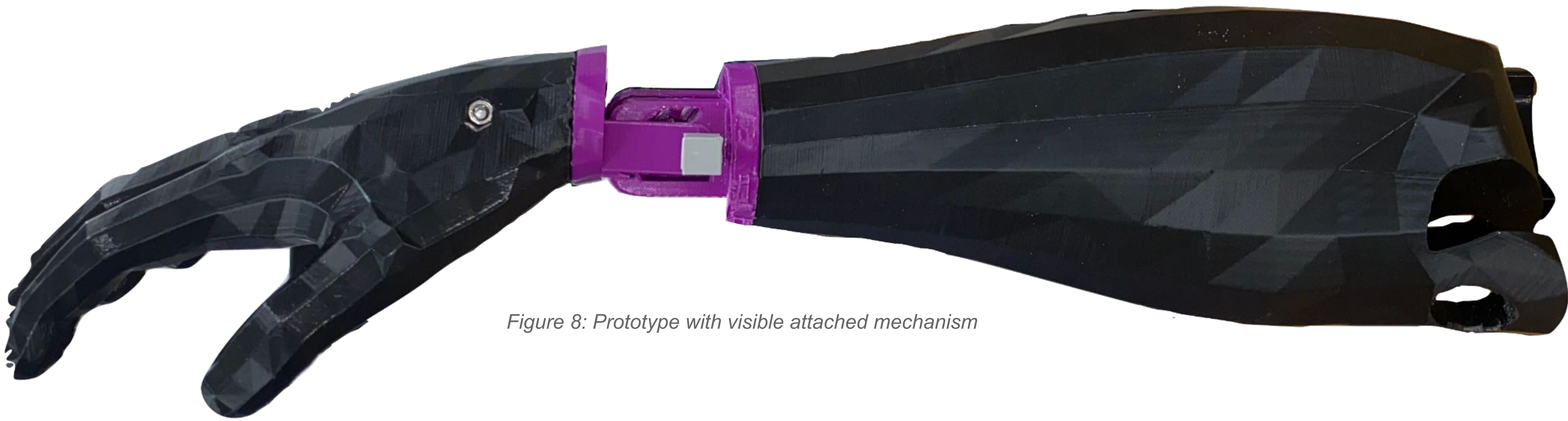


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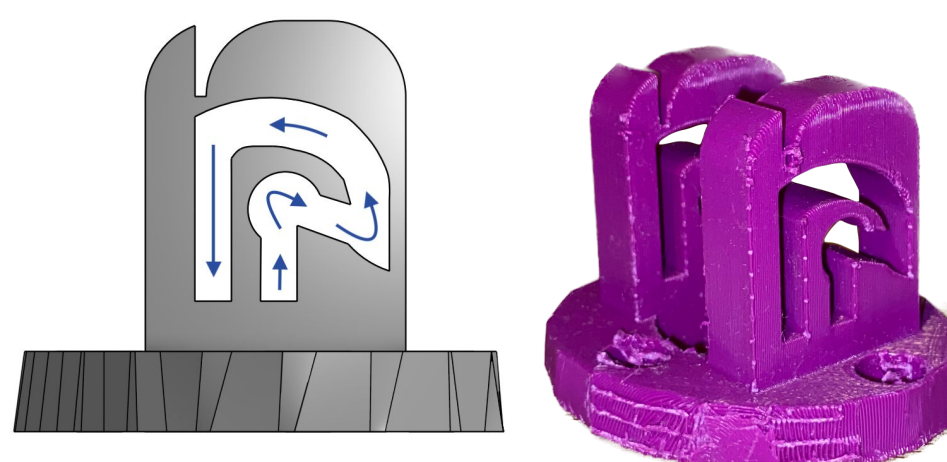


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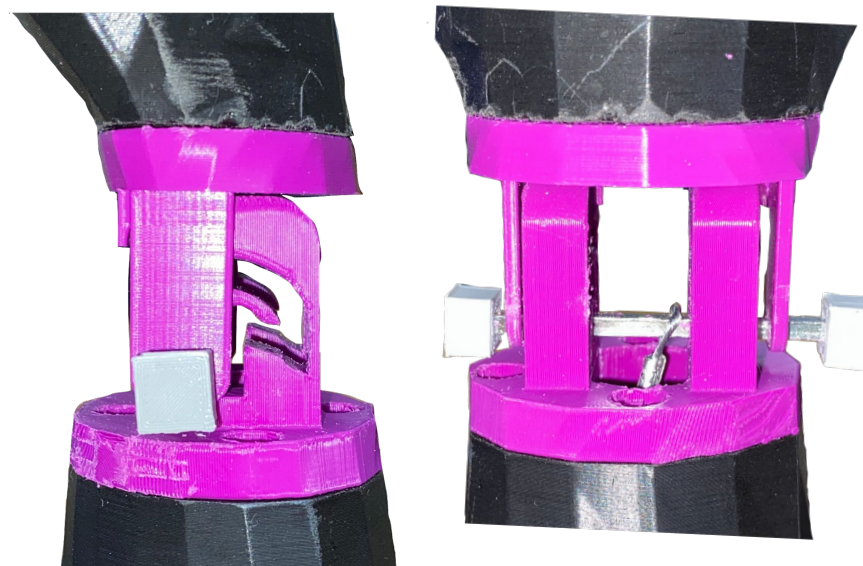


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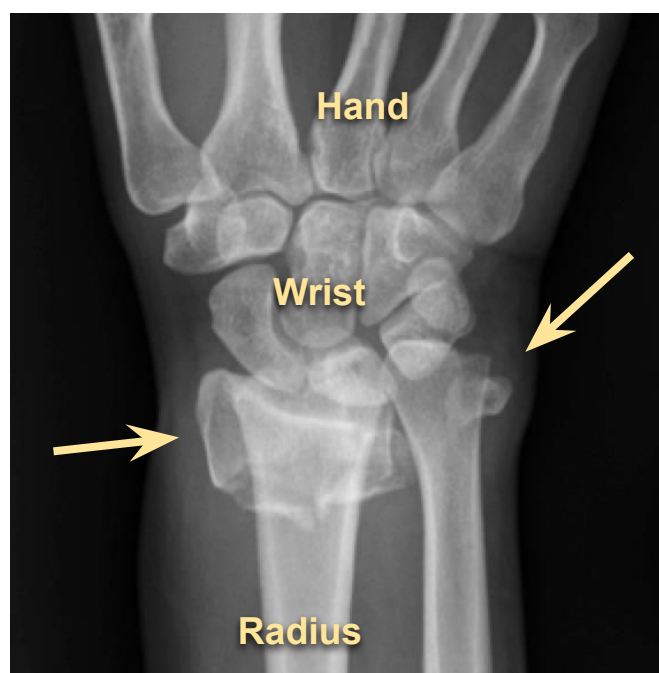


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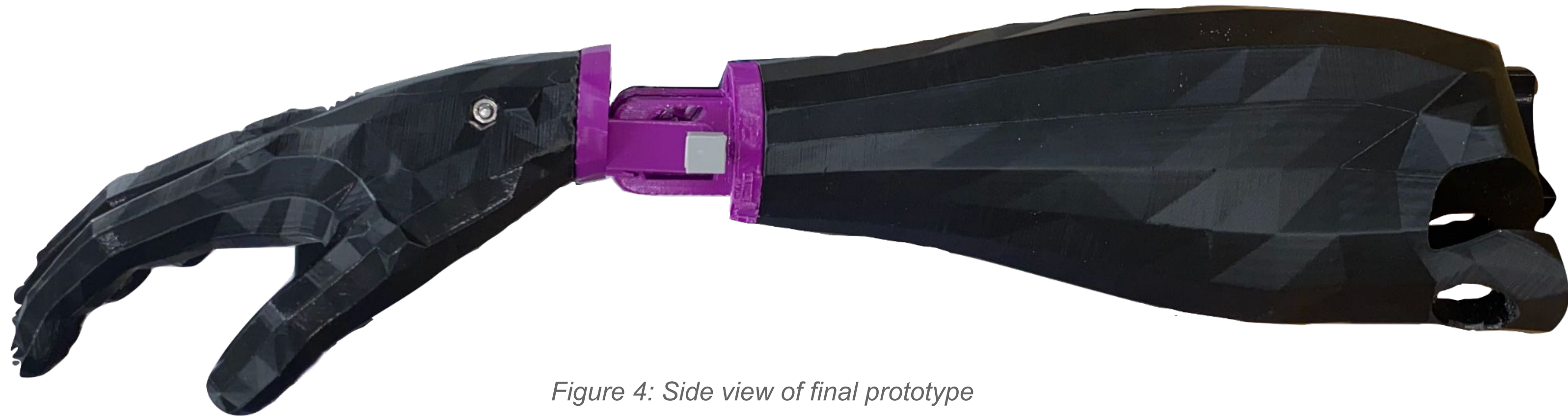


Figure 4: Side view of final prototype

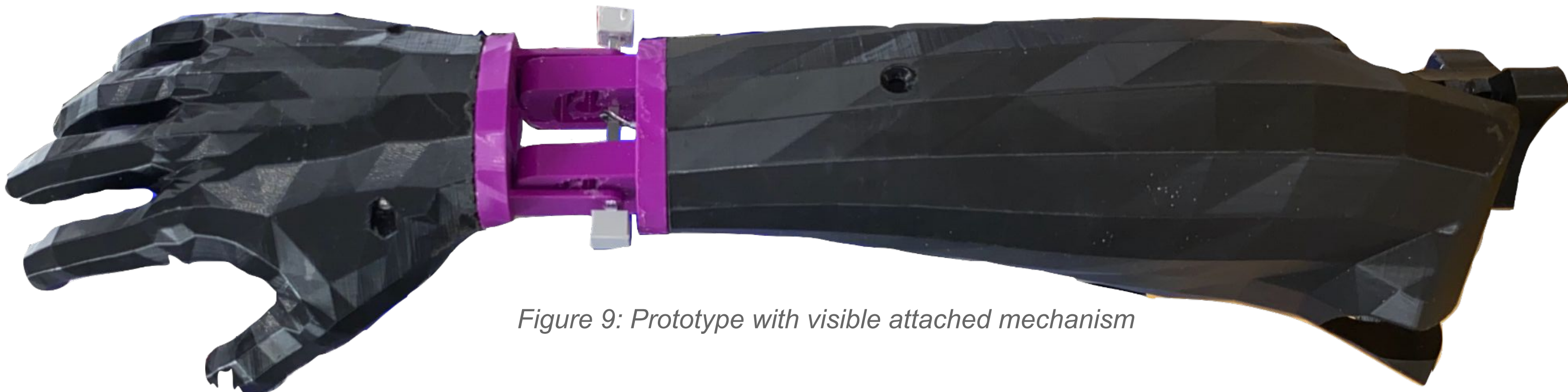


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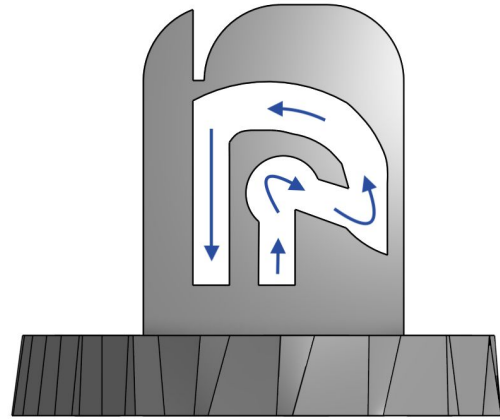


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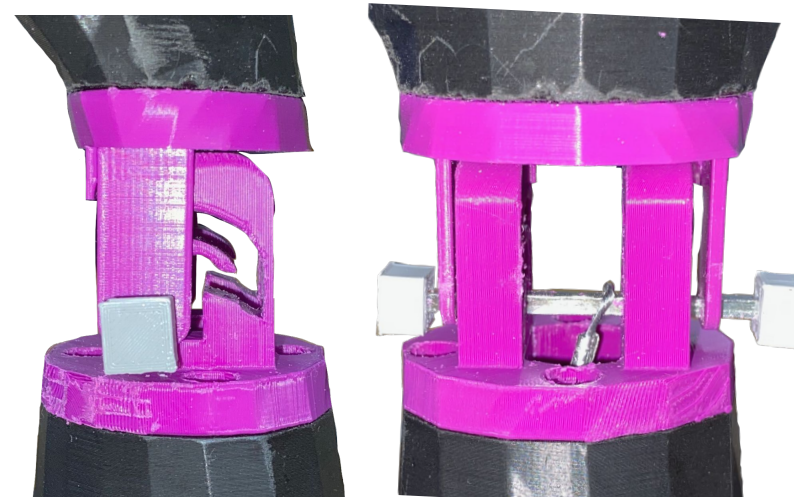


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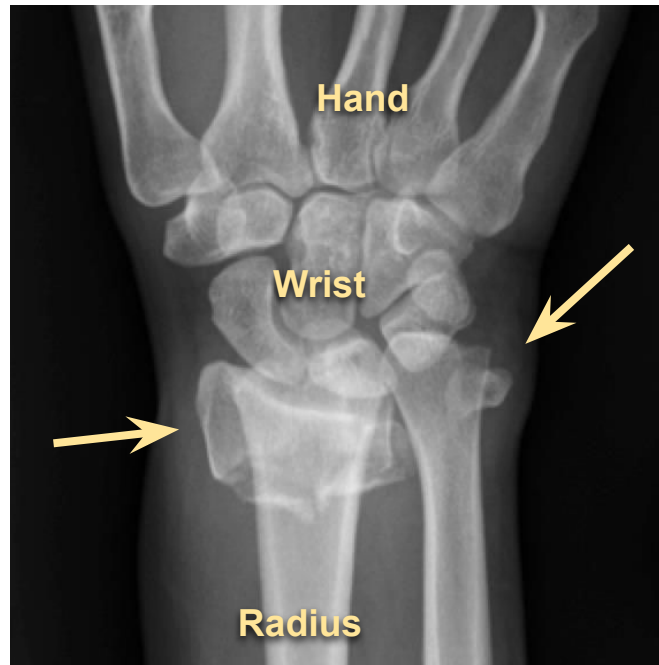


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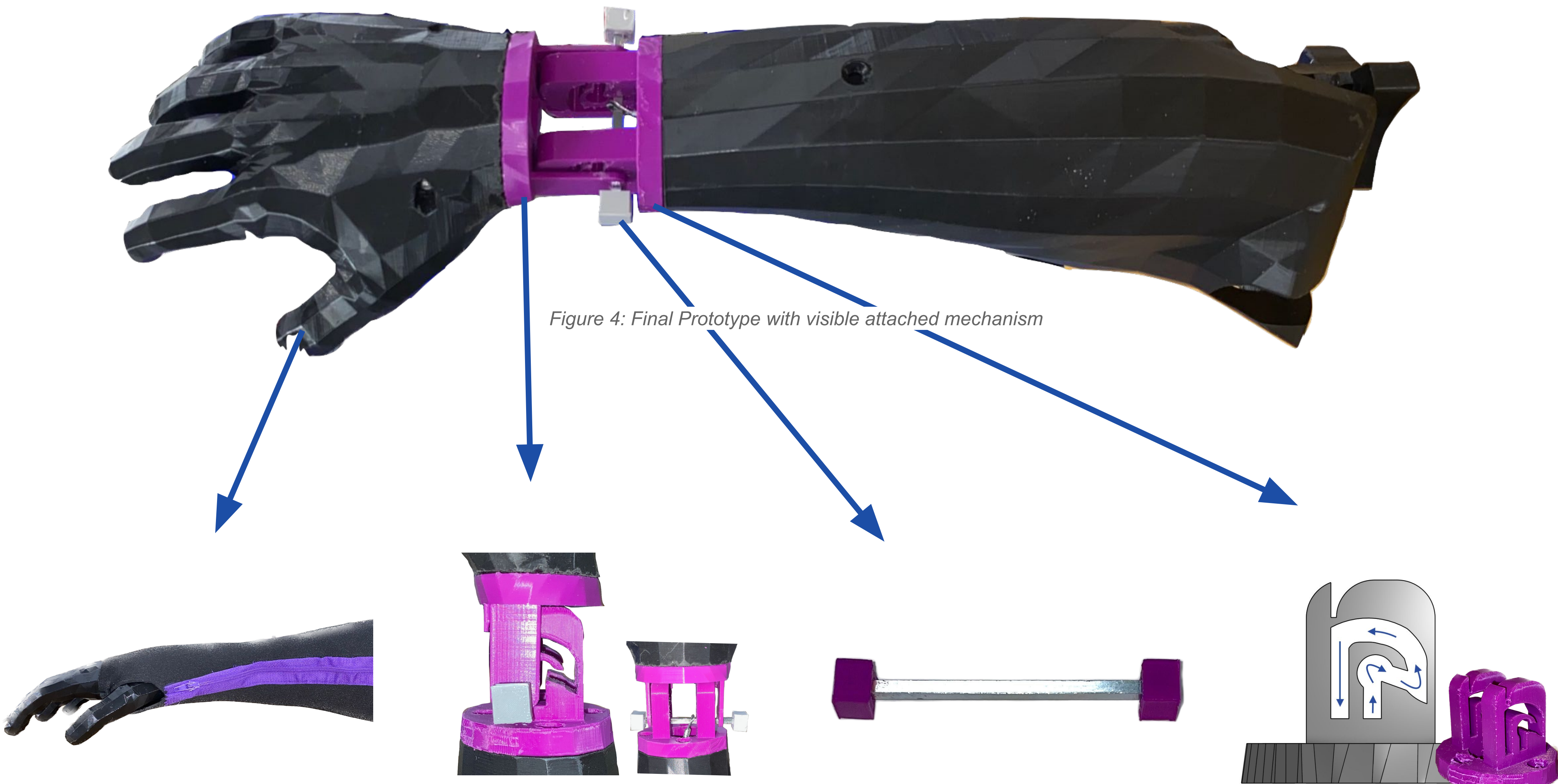


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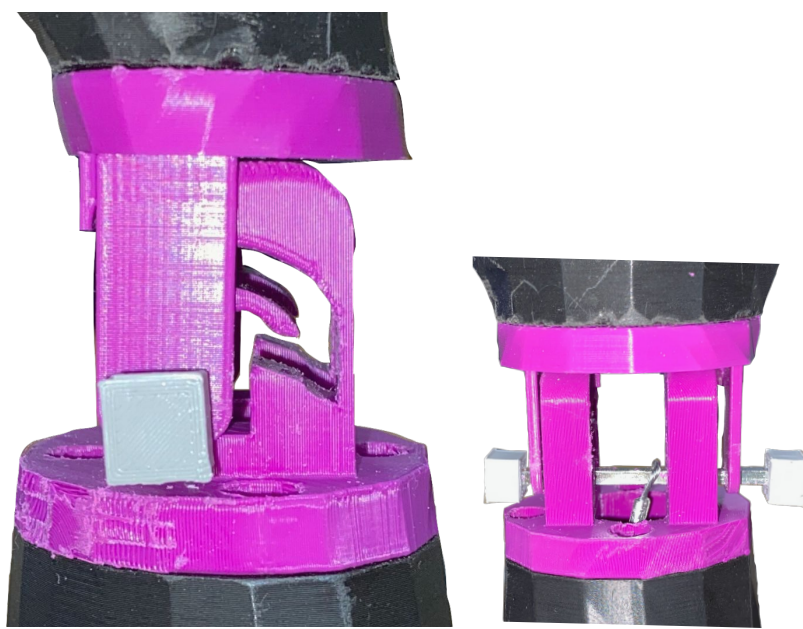


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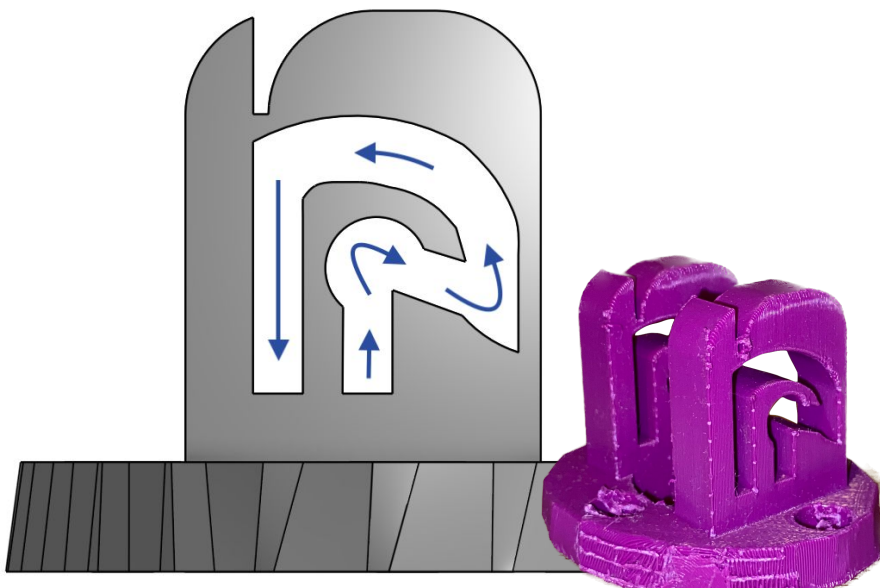


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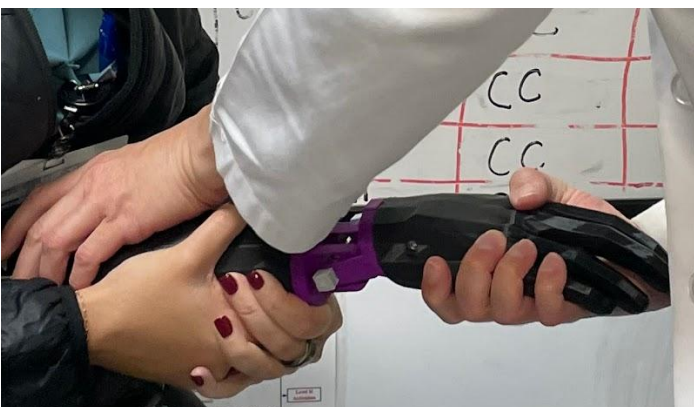


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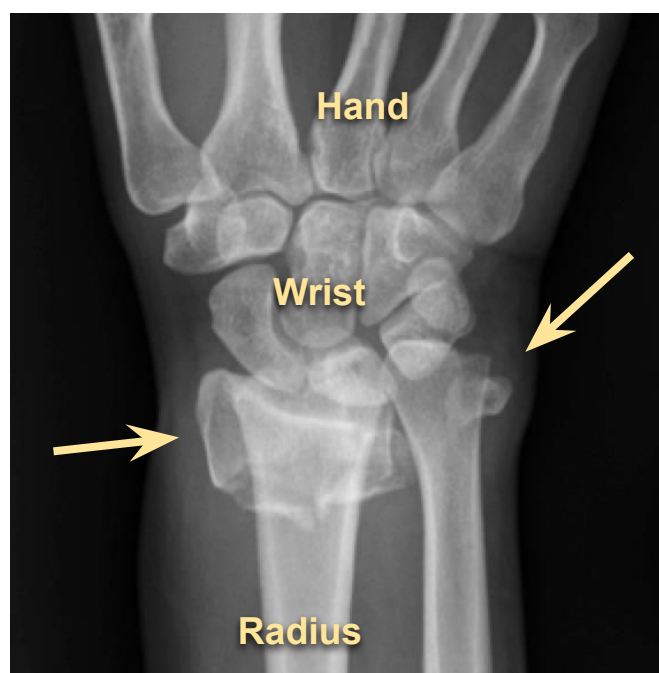


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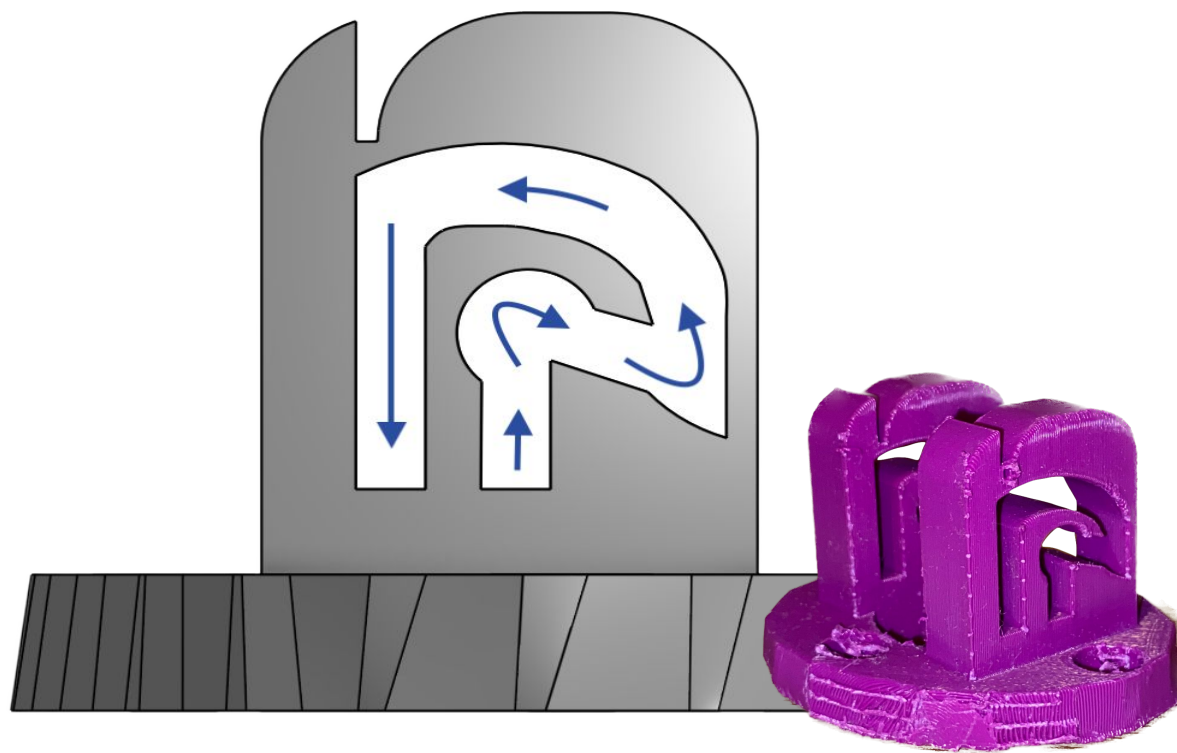


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- Moves along the maze path walls
- Attached to a tensioned wire

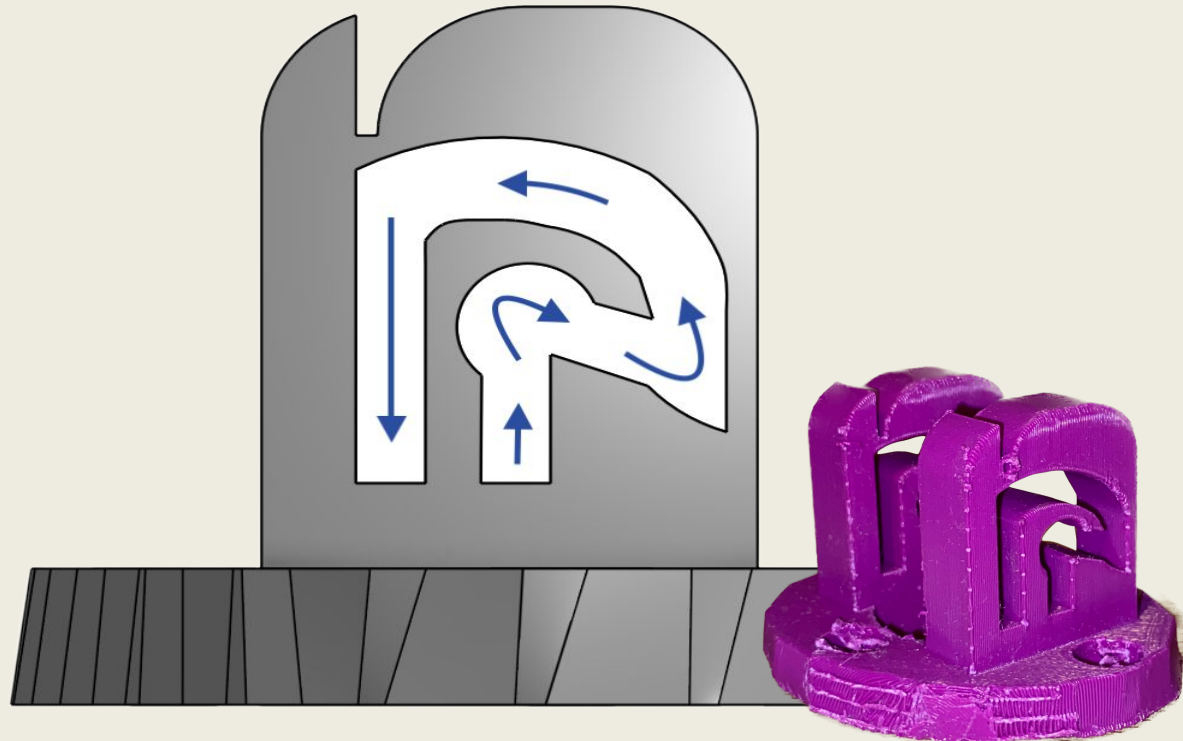


Figure 6: Maze mechanism

- Maze path walls
- Enforces the correct sequences of motion

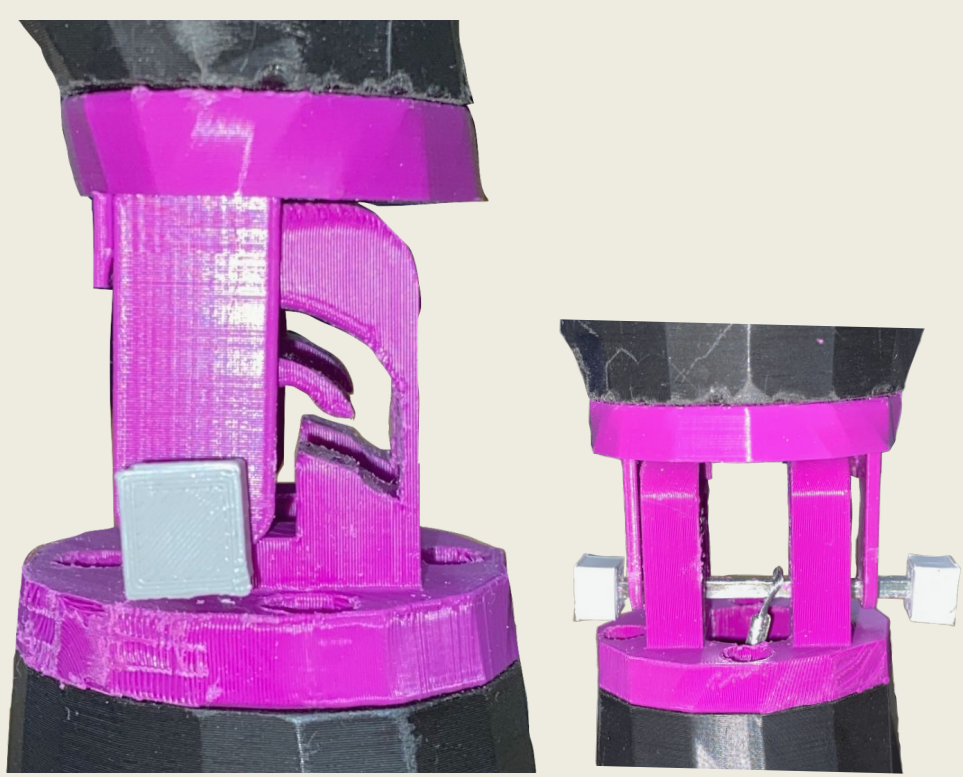


Figure 7: Mechanism in detail

- Piece attached to the hand pulls the rod
- Rod goes through maze path walls
- Wire pulls rod



Figure 8: Prototype covered by neoprene sleeve

- Removable neoprene sleeve
- Improves realism by skin-like feel
- Covers fracture zone

Figure 4: Final Prototype with visible attached mechanism

## Design Criteria and Testing

| Design Criteria             | Procedure                                                                                                                                                     | Target Value                                                   | Results                                    |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------|
| Accurate Mechanism & Forces | 3 physicians performed 10 reductions each using the arm and be asked to rate the force and reduction movements on a 10-point Likert Scale.                    | $\geq 8.0/10$ Average Reduction<br>$\geq 8.0/10$ Average Force | Mechanism: Pass (8.3)<br>Force: Pass (8.1) |
| Safety                      | 20 students used the device and report on the safety perceptions on a 5-point Likert scale.                                                                   | $\geq 4.5/5$ Average                                           | Pass (4.9)                                 |
| Durability                  | The full process of reduction and reset were repeated by team members, students, and physicians to assess the frequency or ease of damage on a 3-point scale. | $\geq 2.8/3$ Average                                           | Pass (2.8)                                 |
| Easy Reset                  | 20 undergraduate students were asked to reset the reduced arm and timed, in addition to answering a user-defined scale.                                       | $< 5$ seconds<br>$\geq 2.5/3$ Average                          | Fail (10.3 seconds)<br>Pass (2.5)          |
| Realistic                   | 35 blindfolded people evaluated the look and feel of the model and rated the material's realism on a 5-point Likert scale.                                    | $\geq 3.5/5$ Average                                           | Fail (3.1)                                 |

## Conclusion & Future Work

The team's design was successful, achieving its main objective, in addition to the approval of the mechanism and forces by physicians. However, there remains room for improvement as the design did not successfully pass all tests.

Moving forward the group aims to:

- Design an automated mechanism for force calibration and alternation.
- Make the model more realistic with materials like silicone.
- Increase the durability by creating a titanium maze (rather than plastic).

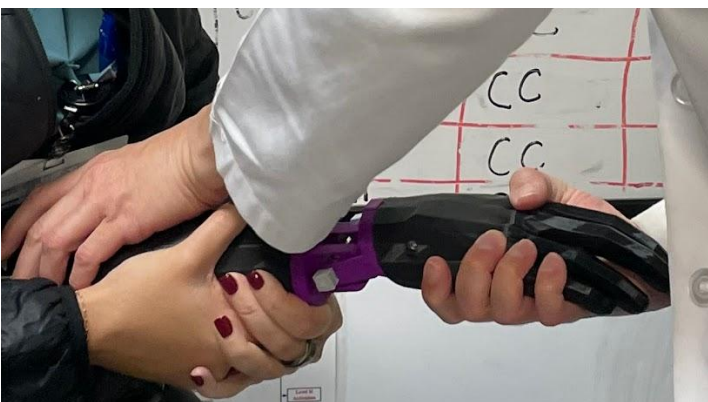


Figure 9: Final prototype being used by resident at the Duke Hospital.

## Acknowledgements & References

Dr. Josh Broder, *Client*  
Dr. Michael Rizk & Dr. Becky Simmons, *Professors*  
David Mercer, *Engineer*  
Ali Stocks, *Lab Manager*  
Kay Lyons, *Teaching Assistant*  
Dr. Jason Luck, *Technical Advisor*  
EGR 101 Section 1

<https://radiopaedia.org/cases/normal-wrist-x-rays>  
<https://www.orthobullets.com/trauma/1027/distal-radius-fractures>



## Problem Statement

Hospitals lack effective tools to train residents on treating one of the most common wrist fractures, a distal radius fracture (Figures 1 & 2). Problems with existing models discouraging hospital investment:

- Expensive
- Fragile
- Inaccurate

The team aims to develop a realistic, durable, and safe model for training closed reductions at the Duke Hospital.



Figure 1: X-Ray of a healthy wrist.



Figure 2: X-Ray of a fractured distal radius in the wrist.

## Initial Prototype

An existing prototype features a 3D-printed hand and forearm, a spring to replicate forces, and a "step" mechanism to mimic the fracture. Issues to address:

- Unrealistic horizontal movement
- Lack of guidance in the mechanism
- Improve overall realism



Figure 3: Initial prototype created by engineer David Mercer

## Design Criteria and Testing

| Design Criteria             | Procedure                                                                                                                                                     | Target Value                                                   | Results                                    |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------|
| Accurate Mechanism & Forces | 3 physicians performed 10 reductions each using the arm and be asked to rate the force and reduction movements on a 10-point Likert Scale.                    | $\geq 8.0/10$ Average Reduction<br>$\geq 8.0/10$ Average Force | Mechanism: Pass (8.3)<br>Force: Pass (8.1) |
| Safety                      | 20 students used the device and report on the safety perceptions on a 5-point Likert scale.                                                                   | $= 4.5/5$ Average                                              | Pass (4.9)                                 |
| Durability                  | The full process of reduction and reset were repeated by team members, students, and physicians to assess the frequency or ease of damage on a 3-point scale. | $\geq 2.8/3$ Average                                           | Pass (2.8)                                 |
| Easy Reset                  | 20 undergraduate students were asked to reset the reduced arm and timed, in addition to answering a user-defined scale.                                       | $\geq 5$ seconds<br>$\geq 2.5/3$ Average                       | Fail (10.3 seconds)<br>Pass (2.5)          |
| Realistic                   | 35 blindfolded people evaluated the look and feel of the model and rated the material's realism on a 5-point Likert scale.                                    | $\geq 3.5/5$ Average                                           | Fail (3.1)                                 |

## Final Prototype Model

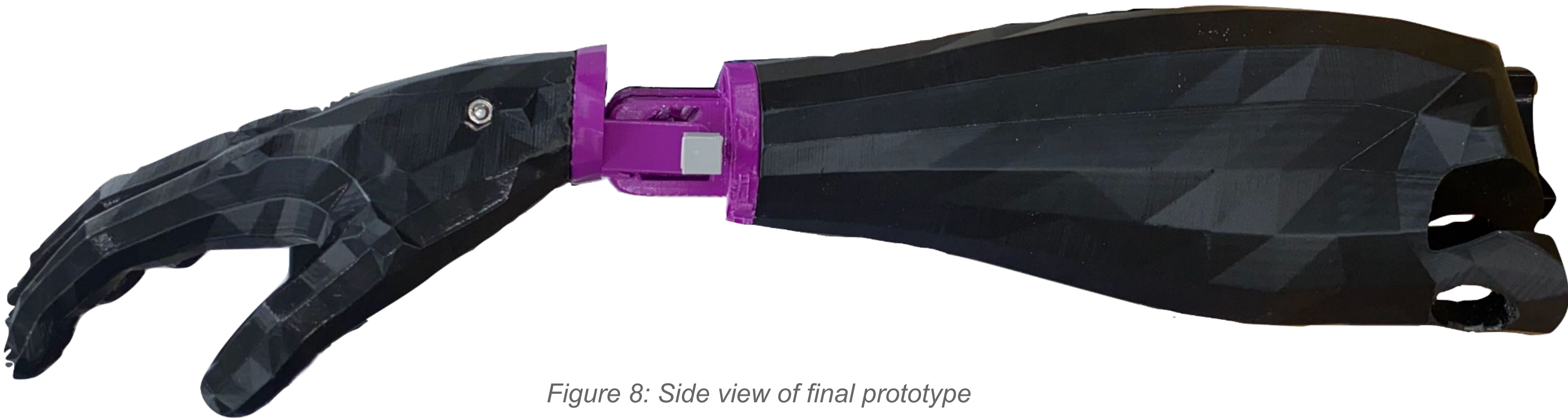


Figure 8: Side view of final prototype

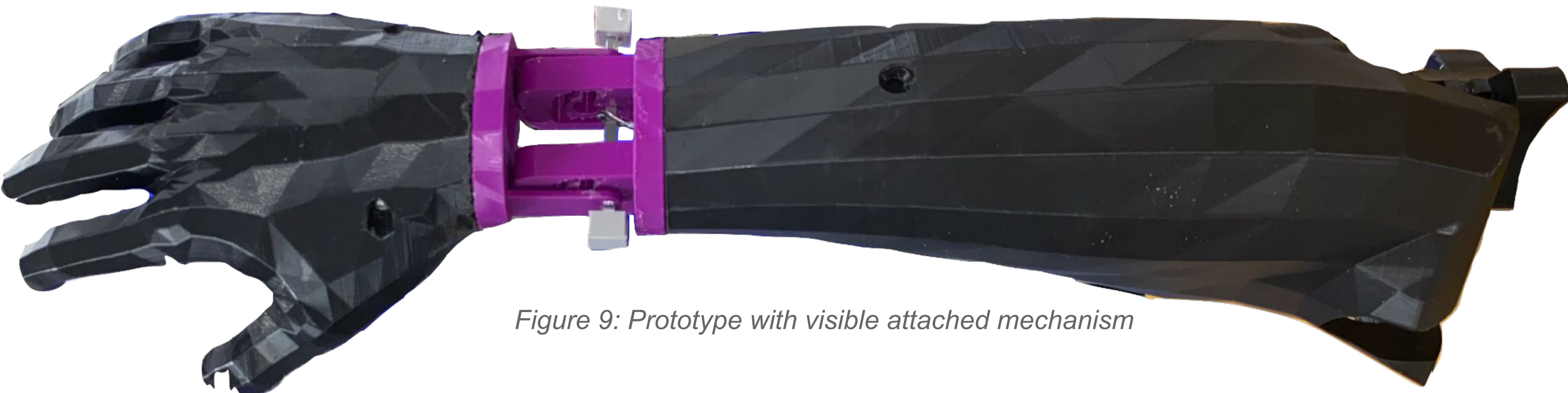


Figure 9: Prototype with visible attached mechanism



Figure 4: Prototype covered by neoprene sleeve



Figure 5: Moving rod

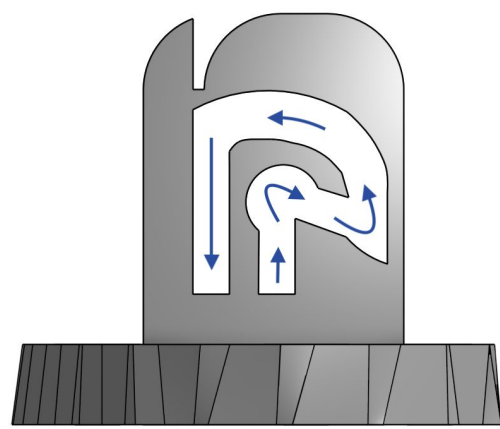


Figure 6: Maze mechanism



Figure 7: Mechanism in detail

- Removable neoprene sleeve
- Improves realism by skin-like feel
- Covers fracture zone

- Square metal rod
- Moves along the path
- Attached to a tensioned wire

- Maze path walls
- Enforces the correct sequences of motion

- Piece attached to the hand pulls the rod
- Rod goes through maze
- Wire pulls rod

## Conclusion & Future Work

The team's design was successful as it achieved its main objective. This can be noticed as as the mechanism and forces were considered to be accurate by physicians. However there is still gap for improvement as the design did not pass every test.

Moving forward the group aims to:

- Establish force markings to adapt the existing prototype
- Make the model more realistic
- Create an easy-reset mechanism
- Increase the durability by creating a titanium maze



Figure 10: Final prototype being used by residents at the Duke Hospital.

## Acknowledgements & References

Client: Dr. Josh Broder  
Professors: Dr. Michael Rizk & Dr. Becky Simmons  
Engineer: David Mercer  
Lab Manager: Ali Stocks  
Teaching Assistant: Kay Lyons  
Advisor: Dr. Jason Luck  
EGR 101 Section 001

<https://radiopaedia.org/cases/normal-wrist-x-rays>  
<https://www.orthobullets.com/trauma/1027/distal-radius-fractures>



# 4Arms – Colles Distal Forearm Fracture Reduction Model

A. He, C. Hanby, C. Gillard, R. Rohr

Pratt School of Engineering, Duke University, Duke School of Medicine



## Problem Statement

Hospitals lack effective training tools for teaching resident physicians how to treat Distal Radius Fractures, represented on Figures 1 and 2. Current training models for closed reductions are too expensive, fragile, and inaccurate, discouraging hospitals from investing in them. The team was thus tasked by their client Dr. Josh Border with improving a pre-existing model arm. Our project aims to create a realistic, safe, and durable prototype to train medical resident how to conduct a closed reduction on a Distal Forearm Fracture at the Duke Hospital.



Figure 1: X-Ray of a healthy wrist.



Figure 2: X-Ray of a fractured distal radius in the wrist.

## Initial Prototype



Figure 3: Initial prototype created by engineer David Mercer

The initial prototype by engineer David Mercer featured a 3D-printed hand and forearm, a spring to replicate reduction forces, and a “step” mechanism to imitate the bone fracture. Thus, the team focused on improving aesthetics and, crucially, fixing the mechanism, which allowed unrealistic horizontal movement and lacked the necessary guidance for residents.



Figure 4: Final sleeve model

A removable neoprene sleeve was designed to increase realism and aesthetics, as it replicates the feeling and stretch of skin, as well as covers the fracture zone, replicating real-life closed reduction procedure.

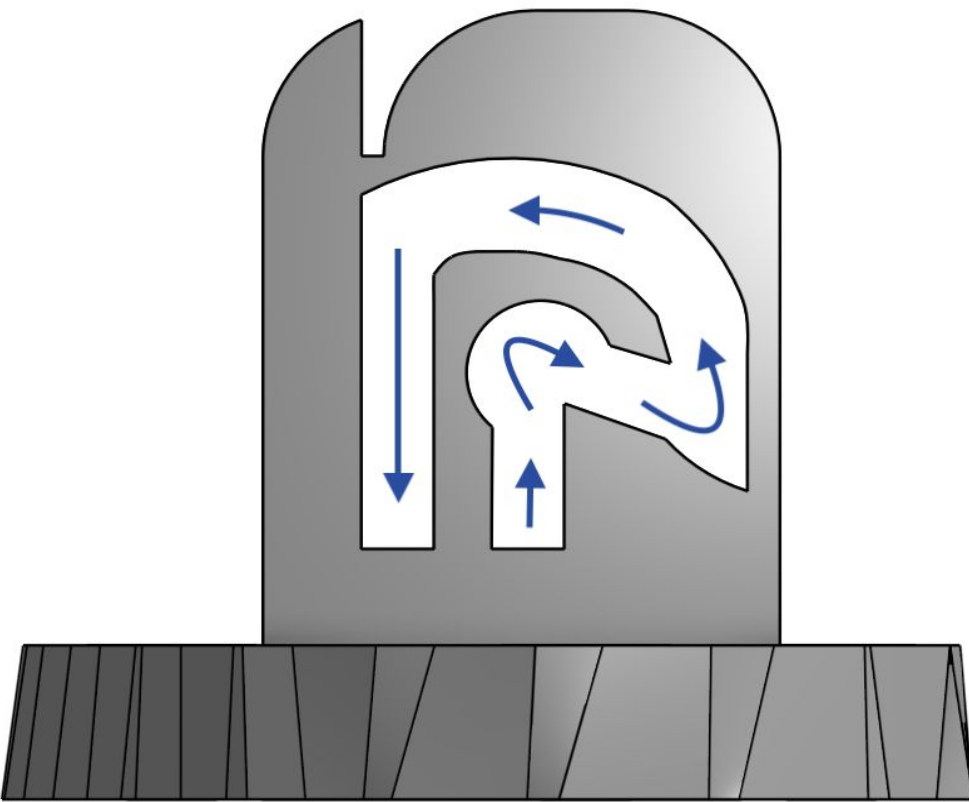


Figure 6: Final path walls

3D-printed path constrains hand movement to replicate a reduction procedure, replacing the two-step process with a guided, step-by-step teaching method.

## Final Prototype Model



Figure 8: Final prototype covered by sleeve

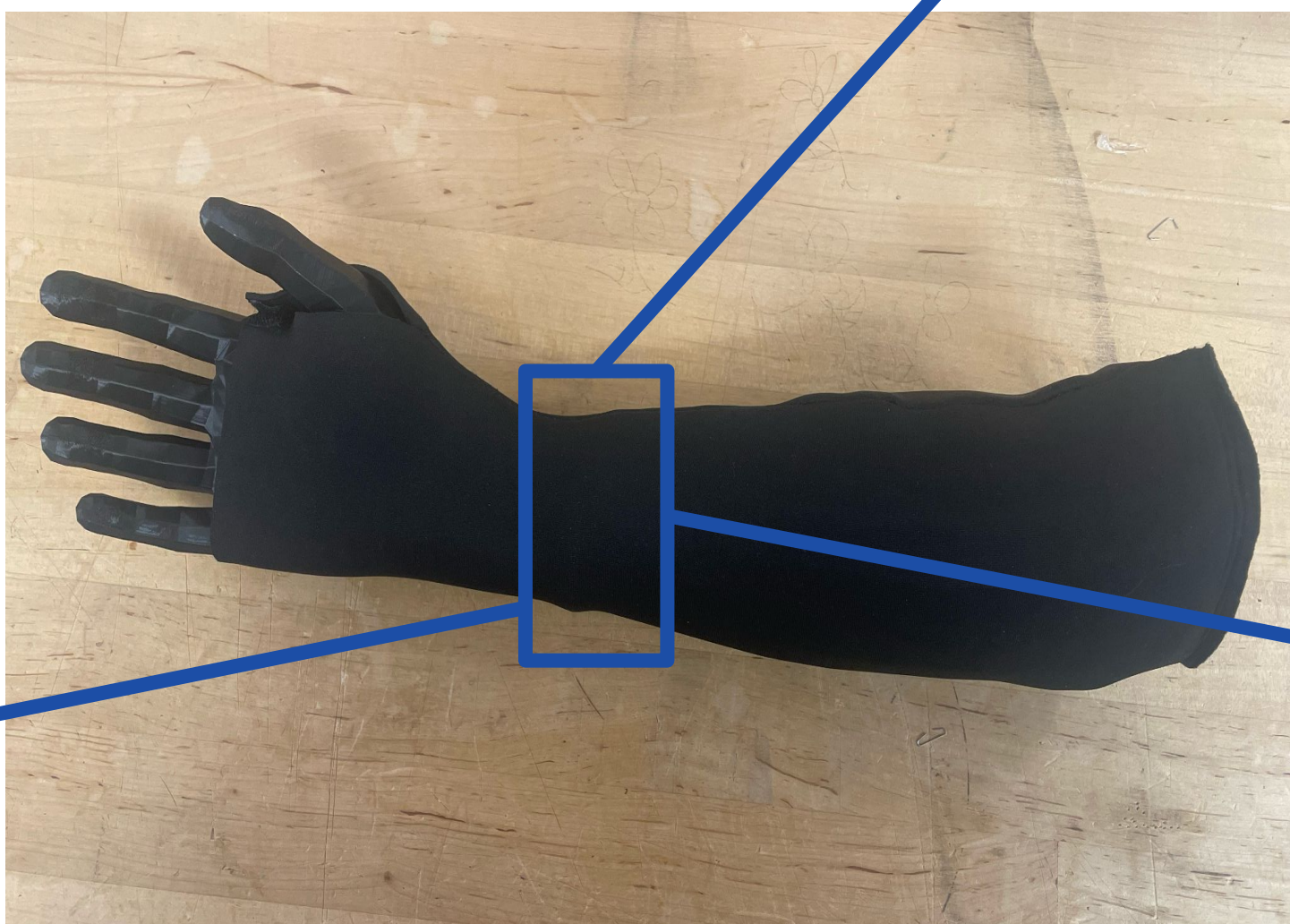


Figure 9: Final prototype with visible attached mechanism

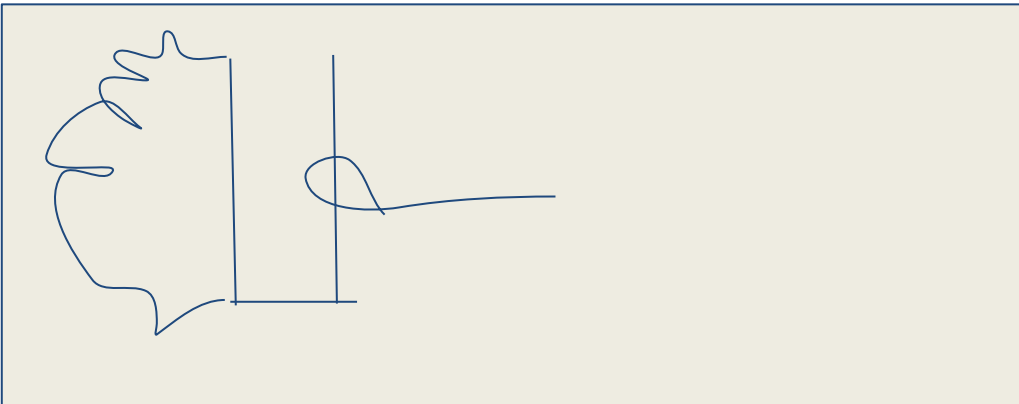


Figure 7: Final mechanism attached

In detail, the moving rod being pulled at the forearm direction by the tensioning wire while it is kept going through the path walls. During a reduction procedure, the performer needs to perform an accurate movement force and direction-wise as to move the rod along its path.



Figure 5: Final moving rod

The metal rod moves through the path during the reduction, and is attached to a tensioned forearm wire, replicating reduction forces. Switching from a round to a square rod resolved the horizontal movement issue, forcing hand angulation.

## Design Criteria and Testing

| Design Criteria             | Procedure                                                                                                                                                     | Target value                                                   | Results                                    |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------|
| Accurate Mechanism & Forces | 3 physicians performed 10 reductions each using the arm and be asked to rate the force and reduction movements on a 10-point Likert Scale.                    | $\geq 8.0/10$ Average Reduction<br>$\geq 8.0/10$ Average Force | Mechanism: Pass (8.3)<br>Force: Pass (8.1) |
| Safety                      | 20 students used the device and report on the safety perceptions on a 5-point Likert scale.                                                                   | = 4.5/5 Average                                                | Pass (4.9)                                 |
| Durability                  | The full process of reduction and reset were repeated by team members, students, and physicians to assess the frequency or ease of damage on a 3-point scale. | $\geq 2.8/3$ Average                                           | Pass (2.8)                                 |
| Easy Reset                  | 20 undergraduate students were asked to reset the reduced arm and timed, in addition to answering a user-defined scale.                                       | $\geq 5$ seconds<br>$\geq 2.5/3$ Average                       | Fail (10.3 seconds)<br>Pass (2.5)          |
| Realistic                   | 35 blindfolded people evaluated the look and feel of the model and rated the material's realism on a 5-point Likert scale.                                    | $\geq 3.5/5$ Average                                           | Fail (3.1)                                 |
|                             |                                                                                                                                                               | $\geq 8.0/10$ Average                                          | Mechanism: Pass (8.3)                      |

## Conclusion & Future Work

The team’s design was \_\_, as it [failed to/achieved] \_\_\_\_\_. This can be noticed as XXXXXXXXXXXX. Some of the positive takeaways are CCCCCCCCCC. Moving forward we hope to:

- Improve the force markings by designing an automatic system to alter the force and calibrate the device.
- Strengthen and increase durability by printing the finalized mechanism with a stronger material such as metal orr..

## Acknowledgements & References

Client: Dr. Josh Broder  
Professors: Dr. Michael Rizk & Dr. Becky Simmons  
Engineer: David Mercer  
Advisor: Dr. Jason Luck  
EGR 101 Section 001

<https://orthoinfo.aaos.org/en/diseases--conditions/distal-radius-fractures-broken-wrist/>



# 4Arms – Colles Distal Forearm Fracture Reduction Model

A. He, C. Hanby, C. Gillard, R. Rohr

Pratt School of Engineering, Duke University, Duke School of Medicine



## Problem Statement

Hospitals lack effective training tools for teaching resident physicians how to treat Distal Radius Fractures, represented on Figures 1 and 2. Current training models for closed reductions are too expensive, fragile, and inaccurate, discouraging hospitals from investing in them. The team was thus tasked by their client Dr. Josh Border with improving a pre-existing model arm, designed by engineer David Mercer. Our project aims to create a locally -sourced, sate, and por



Figure 1: X-Ray of a healthy distal radius



Figure 2: X-Ray of a broken (fractured) distal radius

Initial prototype created by engineer David Mercer consisted of a 3D printed hand and forearm, a spring mechanism to replicate forces of the pseudo-reduction maneuver, and a “step” mechanism to replicate the bone fracture.

Thus, the main issues the team focused on were making it aesthetically realistic, and most importantly, fixing the mechanism which not only allowed for a straight horizontal movement, not constraining it to a realistic reduction motion, but it was not restrictive enough for the residents to guide them.

The initial prototype by engineer David Mercer featured a 3D-printed hand and forearm, a spring to replicate pseudo-reduction forces, and a "step" mechanism for the bone fracture. The team focused on improving aesthetics and, crucially, fixing the mechanism which allowed unrealistic horizontal movement and lacked the necessary guidance for residents.



Figure 3: Initial prototype created by engineer David Mercer

## Initial Prototype

## Design Criteria and Testing

| Design Criteria    | Procedure                                                                                                                                             | Target value                                             | Result |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|--------|
| Accurate mechanism | 3 physicians will perform 10 reductions using the arm and be asked to rate the force and reduction movements on a 10-point Likert Scale.              | $\geq 8/10$ Average                                      |        |
| Accurate forces    | Team used a digital force gauge to check that the forces needed to reduce the injury are accurate.                                                    | $26.3 \pm 2$ lbf<br>$31.1 \pm 2$ lbf<br>$29.3 \pm 2$ lbf |        |
| Safety             | 20 students will use the device and report on the safety perceptions on a 5-point Likert scale                                                        | $= 5/5$ Average                                          |        |
| Durability         | The full process of reduction and reset will be repeated by team members and physicians to assess the frequency or ease of damage on a 3-point scale. | $\geq 2.75/3$ Average                                    |        |
| Easy reset         | 20 medical residents will be asked to reset the reduced arm and timed, in addition to a user defined scale.                                           | $\geq 5$ seconds<br>$\geq 2.75/3$ Average                |        |
| Realistic          | 35 Blindfolded people evaluate the look and feel of the model and rate the material's realism on a 5-point Likert scale.                              | $\geq 3.5/5$ Average                                     |        |

## Final Prototype Model



Figure 4: Final sleeve model

Removable neoprene sleeve that attributes to the arm a realistic aspect, as its texture is similar to skin, as well as covers the fracture zone, replicating real-life procedure situations

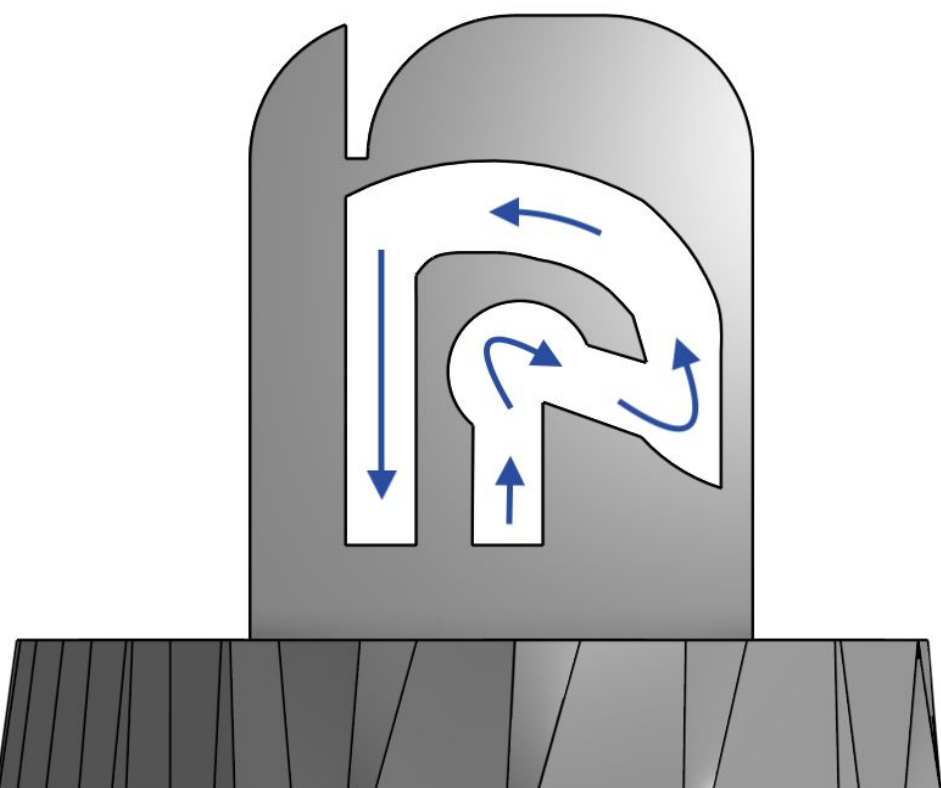


Figure 6: Final path walls

3D-Printed path walls designed to constrain the hand movement as to replicate accurately the motion of a reduction procedure.



Figure 8: Final prototype covered by sleeve

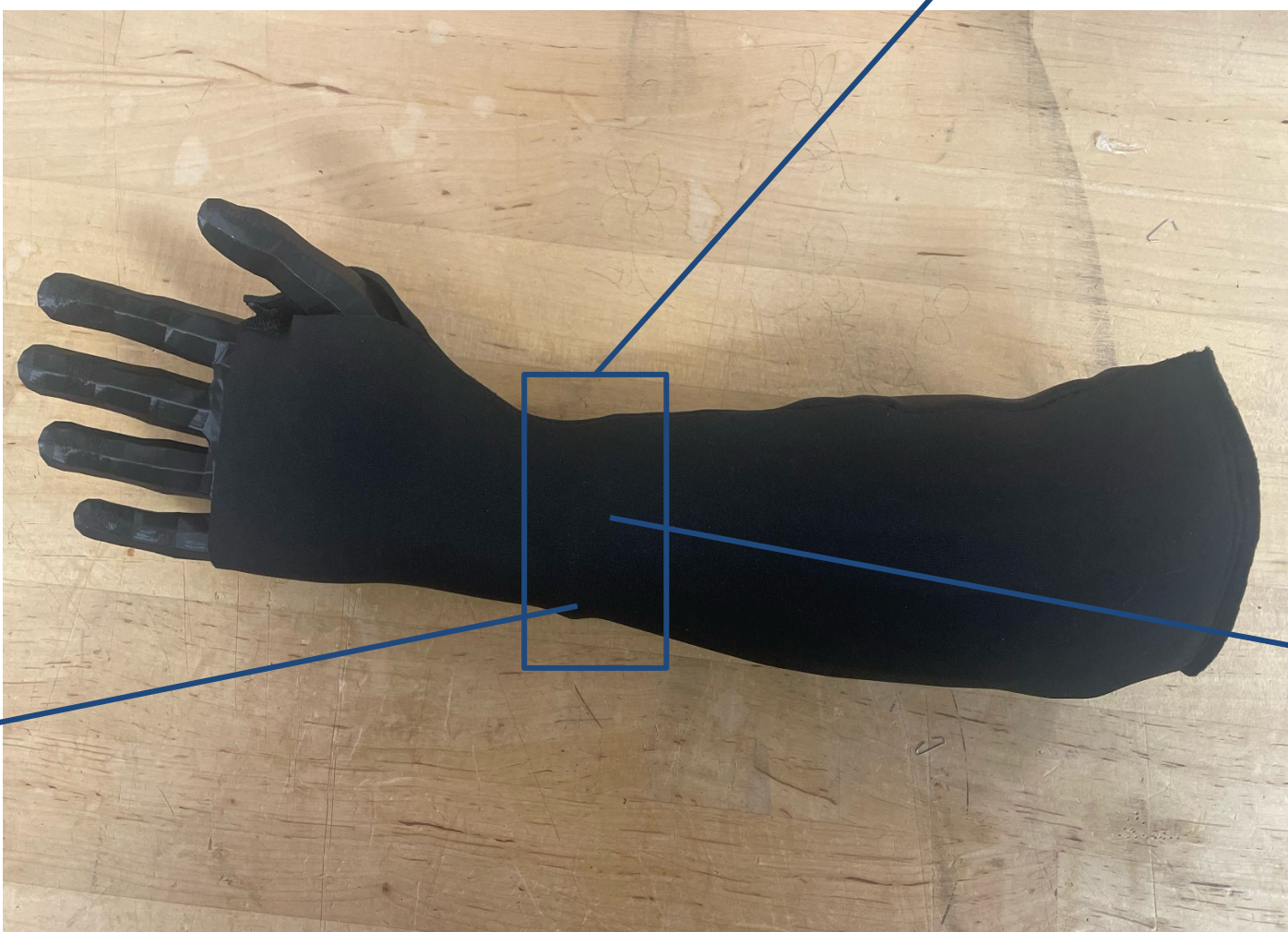


Figure 9: Final prototype with visible attached mechanism

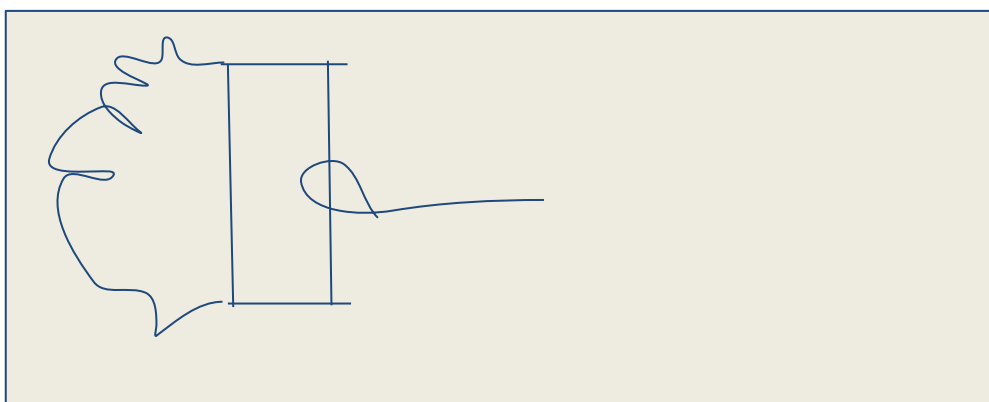


Figure 7: Final mechanism attached

In detail, the moving rod being pulled at the forearm direction by the tensioning wire while it is kept going through the path walls. During a reduction procedure, the performer needs to perform an accurate movement force and direction-wise as to move the rod along its path.



Figure 5: Final moving rod

Metal moving rod that moves through the path walls as a reduction motion is performed while being attached to a tensioned wire at the forearm replicating the force to exert a reduction.

## Conclusion & Future Work

The team’s design was \_\_, as it [failed to/achieved] \_\_\_\_\_. The This can be noticed as XXXXXXXXXXXX. Some of the positive takeaways are CCCCCCCCCC.

## References

Client: Dr. Josh Broder  
Professors: Dr. Michael Rizk & Dr. Becky Simmons  
Engineer: David Mercer  
Advisor: Dr. Jason Luck  
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